

COMPUTER GRAPHICS
GROUP
PROJECT

INTRODUCTION

Computer Graphics is a field that focuses on creating visual content using programming, and is used in games and simulations. This project uses C++ and the graphics.h library to design a space scene containing objects like a spaceship, planets, and stars. It includes two versions: a static scene (no movement) and a dynamic scene (with animation).

The goal is to demonstrate the difference between static and dynamic graphics and how animation is implemented.

METHODOLOGY

The project was developed using a modular programming approach, where each object is created in a separate function to improve organization and readability.

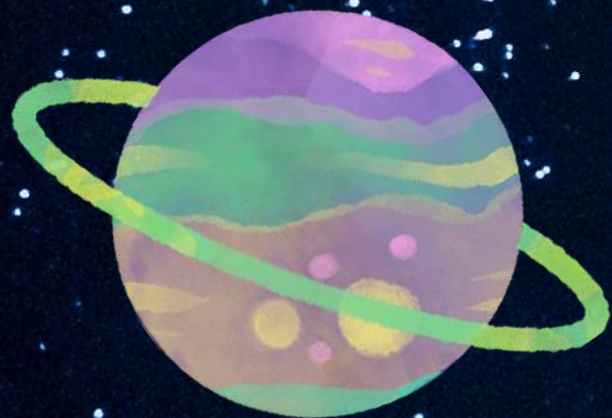
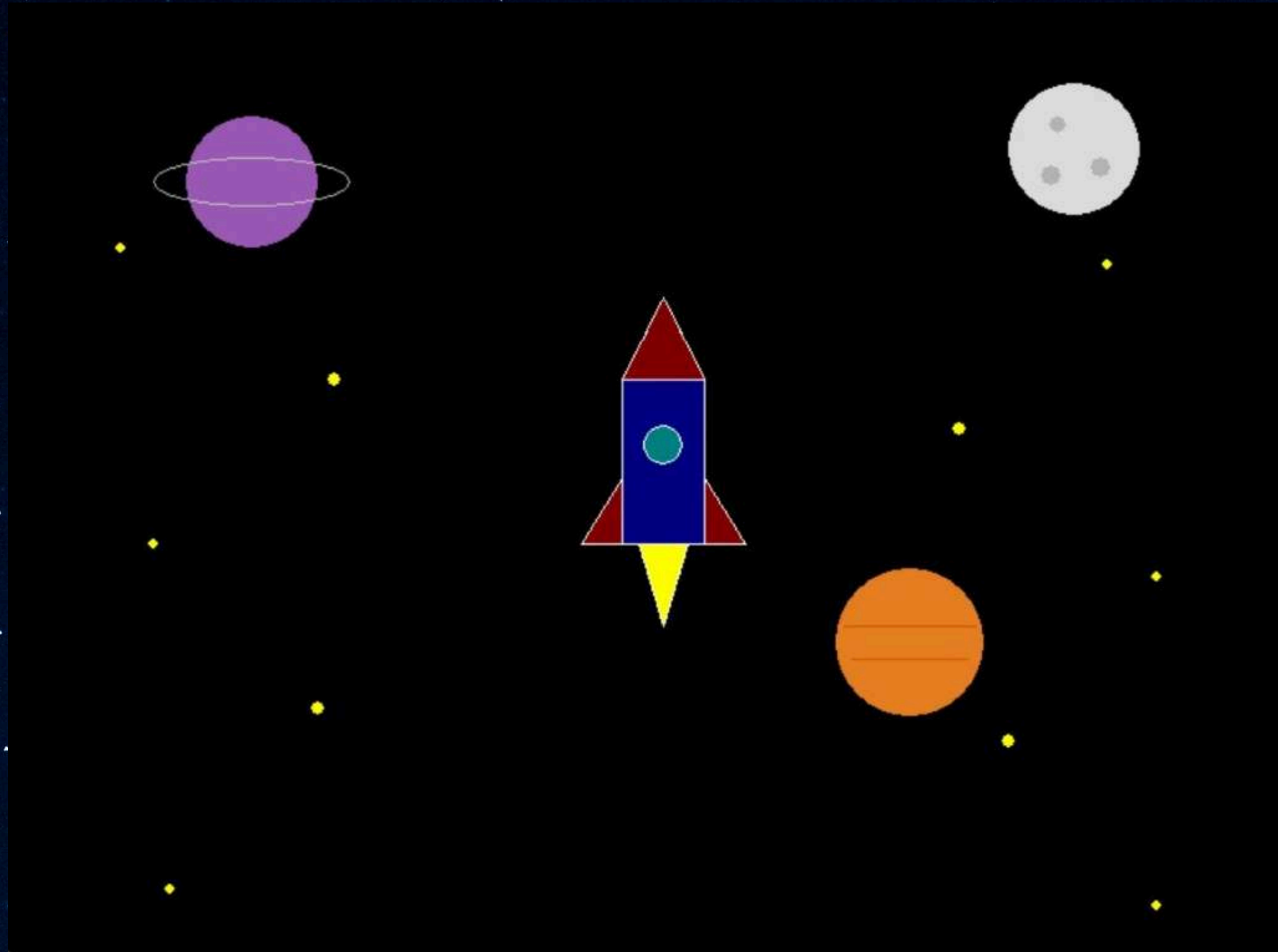
TOOLS USED :

- C++
- graphics.h library
- Mathematical functions (sin, cos)

IMPLEMENTATION :

- Spaceship: Drawn using basic shapes (rectangle, lines, circle) with color and engine flame effect.
- Stars: Small static shapes distributed across the screen.
- Planets: Two planets with colors and details; in the dynamic version they move in circular motion using sin and cos.
- Moon: Large shape with craters placed in the top-right corner.

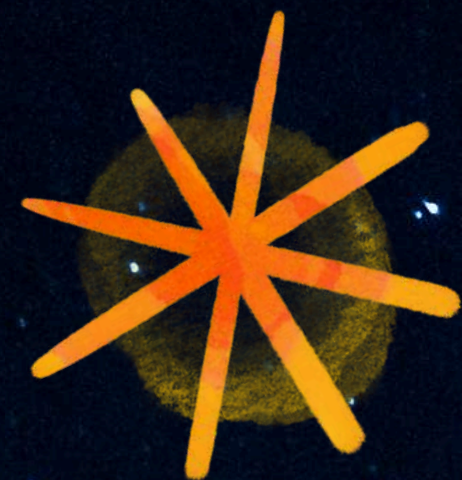
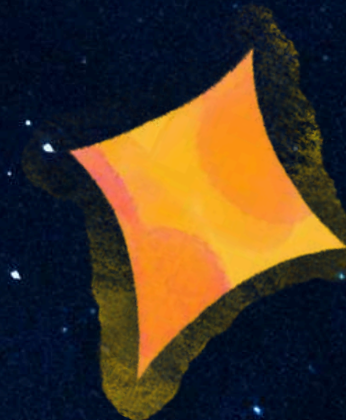
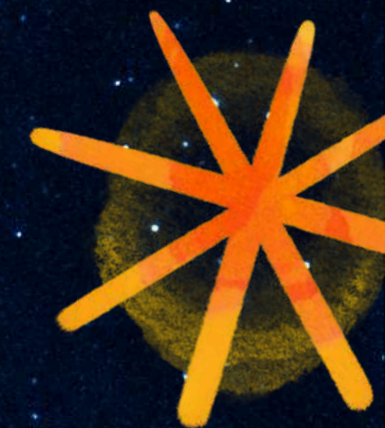
STATIC SCENE



DYNAMIC SCENE



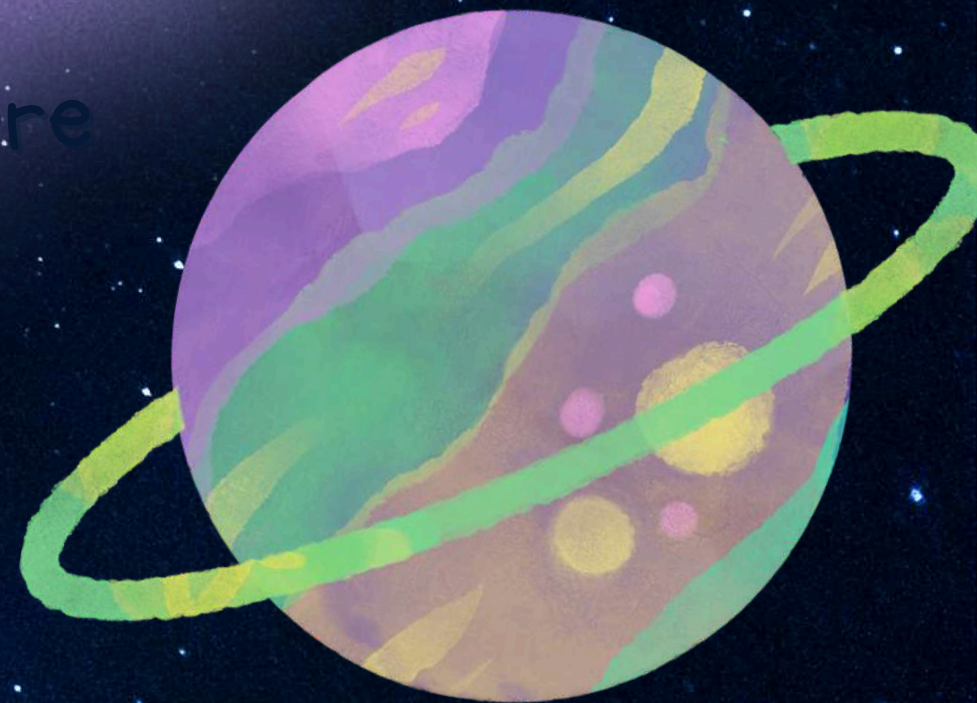
CODE AND DEMO



CONCLUSION

The project successfully created a space scene in both static and dynamic forms using C++ and graphics.h , It demonstrates drawing shapes, coordinate handling, and animation using loops and mathematical functions like sin and cos ,

Overall, it provided practical experience in computer graphics and a foundation for more advanced applications.



THANK
YOU

